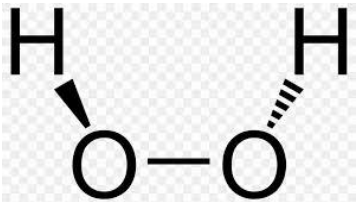
Inorganic Cumulative Exam
Feb 7 2020

Answer any 5 of the following 10 questions (20 points each). Any answers provided beyond the requested five will not be graded. Remember: Your answers to these questions are meant to provide evidence of broad, graduate-school level chemistry knowledge. Consider this in your choice of terminology and examples as answers to these questions.

1. Use your inorganic/materials chemistry knowledge to describe the basic operating principle of an automobile's catalytic converter, including the materials involved and two of the reactions catalyzed in the device.



2. Given the molecular structure of H_2O_2 and assorted character tables:
   a. (4 pts) What is the point group of H_2O_2?
   b. (10 pts) Assume that you need to prove that H_2O_2 has the shown structure and is not planar. Without knowing the energies of the IR absorption modes, use group theory to demonstrate if you can discriminate between the given structure and a planar (trans) structure with IR spectroscopy? (Hint: you can assume that each of the IR-active modes are distinguishable in your spectrometer).
   c. (6 pts) Given the same question as (b), can you discriminate between the two conformations with Raman spectroscopy?

3.
   a. (15 pts) Using a frontier orbital diagram, describe how a Jahn-Teller distortion stabilizes a generic ML_6 complex (M= metal cation, L = sigma-bonding ligand).
   b. (5 pts) List the d-electron count(s) and spin configurations for which a JT distortion does not improve the stability of the complex over the octahedral complex.

4.
   a. (10 pts) Define hydration isomerism.
   b. (10 pts) Give an example of a compound and two of its hydration isomers.

5.

a. (7pts) Do you expect the H_3^+ cation to be linear or triangular? Explain your answer?
b. (7 pts) Describe an experiment that could be used to answer (a), including specific details about how the proposed measurement would discriminate between the two.
c. (6 pts) Given your answer to (a), sketch the electron density of the HOMO level of the H_3^+ cation.

6.

a. (10 pts) Define hydrogen bonding, including a description of the electronic and atomic structure of the participating molecules.
b. (10 pts) List four examples of hydrogen bond donor/acceptor pairs.

7. An exciton is the bound state of an excited electron-hole pair in a material (by photon absorption or by other means).

a. (10 pts) How does the dielectric constant of the material (ε) affect the fate of the exciton (lifetime, recombination, dissociation, etc.)
b. (10 pts) Why is an exciton a good model for photoexcited carriers in biomolecular light absorbers such as photosystem II?

8. Other than carbon, name two other main group elements that can form polymers and describe the structure of an example polymer for each element.

9.

a. (6 pts) Define the term lattice energy in the context of crystal formation.
b. (4 pts) Is the term most appropriate for covalent network solids or ionic solids. Explain your rationale in terms of atomic and electronic interactions in each of these classes of solid.
c. (10 pts) How does the lattice energy of a solid relate to its solubility (in water, for example)? Use Hess's law to justify your answer.

10.

a. (6 pts) Assume you are given a simple cubic lattice (lattice constant a = 0.3 nm), what is the atop surface atom density of the (hkl) = (1,0,0) facet?
b. (6 pts) What is that atop surface atom density for the (1,1,1)?
c. (8 pts) Sketch the lattice plane(s) cutting through the unit cell for the (hkl) = (1,1,0) Bragg planes in the simple cubic lattice.

$$\# \, reps = \frac{1}{h}\sum(g * \chi_r * \chi_i)$$

h = total operations in a group

g = number of operations in a class

χ_r = the character of the reducible representation

χ_i= the character of the irreducible representation

The transformation matrix for a C_n rotation around the z axis is

$$\begin{bmatrix} cos\,\theta & -sin\theta & 0 \\ sin\theta & cos\,\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$cos\frac{2\pi}{1} = 1, sin\frac{2\pi}{1} = 0$

$cos\frac{2\pi}{2} = -1, sin\frac{2\pi}{2} = 0$

$cos\frac{2\pi}{3} = -0.5, sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$

$cos\frac{2\pi}{4} = 0, sin\frac{2\pi}{4} = 1$

$cos\frac{2\pi}{5} = \frac{1}{4}(-1+\sqrt{5}), sin\frac{2\pi}{5} = \frac{1}{4}\sqrt{10+2\sqrt{5}}$

$cos\frac{2\pi}{6} = 0.5, sin\frac{2\pi}{6} = \frac{\sqrt{3}}{2}$

Character Tables

C_2	E	C_2		
A	1	1	z, R_z	x^2, y^2, z^2, xy
B	1	-1	x, y, R_x, R_y	yz, xz

C_3	E	C_3	$(C_3)^2$		
A	1	1	1	z, R_z	x^2+y^2, z^2
E	1 1	e e^*	e^* e	x+iy; R_x+iR_y x-iy; R_x-iR_y	(x^2-y^2, xy) (yz, xz)

e = exp(2πi/3)

C_4	E	C_4	C_2	$(C_4)^3$		
A	1	1	1	1	z, Rz	x^2+y^2, z^2
B	1	-1	1	-1		x^2-y^2, xy
E	1 1	i -i	-1 -1	-i i	x+iy; R_x+iR_y x-iy; R_x-iR_y	(yz, xz)

C_{2v}	E	C_2	$\sigma_v(xz)$	$\sigma_v(yz)$		
A_1	1	1	1	1	z	x^2, y^2, z^2
A_2	1	1	-1	-1	R_z	xy
B_1	1	-1	1	-1	x, R_y	xz
B_2	1	-1	-1	1	y, R_x	yz

C_{3v}	E	$2C_3$	$3\sigma_v$		
A_1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	-1	R_z	
E	2	-1	0	(x, y) (R_x, R_y)	(x^2-y^2, xy) (xz, yz)

C_{4v}	E	$2C_4$	C_2	$2\sigma_v$	$2\sigma_d$		
A_1	1	1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	1	-1	-1	R_z	
B_1	1	-1	1	1	-1		x^2-y^2
B_2	1	-1	1	-1	1		xy
E	2	0	-2	0	0	(x, y) (R_x, R_y)	(xz, yz)

C_{6v}	E	$2C_6$	$2C_3$	C_2	$3\sigma_v$	$3\sigma_d$		
A_1	1	1	1	1	1	1	z	x2+y2, z2
A_2	1	1	1	1	-1	-1	Rz	
B_1	1	-1	1	-1	1	-1		
B_2	1	-1	1	-1	-1	1		
E_1	2	1	-1	-2	0	0	(x, y) (Rx, Ry)	(xz, yz)
E_2	2	=1	=1	2	0	0		(x^2-y^2, xy)

C_{5v}	E	$2C_5$	$2(C_5)^2$	$5\sigma_v$		
A_1	1	1	1	1	z	x2+y2, z2
A_2	1	1	1	-1	Rz	
E_1	2	2cos(2π/5)	2cos(4π/5)	0	(x, y) (Rx, Ry)	(xz, yz)
E_2	2	2cos(4π/5)	2cos(2π/5)	0		(x2-y2, xy)

D_{2h}	E	C_2 (z)	C_2 (y)	C_2 (x)	i	σ (xy)	σ (xz)	σ (yz)		
A_g	1	1	1	1	1	1	1	1		x^2, y^2, z^2
B_{1g}	1	1	-1	-1	1	1	-1	-1	R_z	xy
B_{2g}	1	-1	1	-1	1	-1	1	-1	R_y	xz
B_{3g}	1	-1	-1	1	1	-1	-1	1	R_x	yz
A_u	1	1	1	1	-1	-1	-1	-1		
B_{1u}	1	1	-1	-1	-1	-1	1	1	z	
B_{2u}	1	-1	1	-1	-1	1	-1	1	y	
B_{3u}	1	-1	-1	1	-1	1	1	-1	x	

D_{3h}	E	$2C_3$	$3C'_2$	σ_h	$2S_3$	$3\sigma_v$		
A'_1	1	1	1	1	1	1		x^2+y^2, z^2
A'_2	1	1	-1	1	1	-1	R_z	
E'	2	-1	0	2	-1	0	(x, y)	(x^2-y^2, xy)
A''_1	1	1	1	-1	-1	-1		
A''_2	1	1	-1	-1	-1	1	z	
E''	2	-1	0	-2	1	0	(R_x, R_y)	(xz, yz)

D_{4h}	E	$2C_4$	C_2	$2C'_2$	$2C''_2$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$		
A_{1g}	1	1	1	1	1	1	1	1	1	1		x^2+y^2, z^2
A_{2g}	1	1	1	-1	-1	1	1	1	-1	-1	Rz	
B_{1g}	1	-1	1	1	-1	1	-1	1	1	-1		x^2-y^2
B_{2g}	1	-1	1	-1	1	1	-1	1	-1	1		xy
E_g	2	0	-2	0	0	2	0	-2	0	0	(Rx, Ry)	(xz, yz)
A_{1u}	1	1	1	1	1	-1	-1	-1	-1	-1		
A_{2u}	1	1	1	-1	-1	-1	-1	-1	1	1	z	
B_{1u}	1	-1	1	1	-1	-1	1	-1	-1	1		
B_{2u}	1	-1	1	-1	1	-1	1	-1	1	-1		
E_u	2	0	-2	0	0	-2	0	2	0	0	(x, y)	

T_d	**E**	**$8C_3$**	**$3C_2$**	**$6S_4$**	**$6\sigma_d$**		
A_1	1	1	1	1	1		$x^2+y^2+z^2$
A_2	1	1	1	-1	-1		
E	2	-1	2	0	0		$(2z^2-x^2-y^2, x^2-y^2)$
T_1	3	0	-1	1	-1	(R_x, R_y, R_z)	
T_2	3	0	-1	-1	1	(x, y, z)	(xy, xz, yz)